

Positioning, Impact, and Applications

Similar experiments in the literature, what is unique in this project, and where linear-map adapters save time, energy, and unlock local models

1. Similar experiments and systems: the landscape

The project sits at the intersection of two research lines — representation-convergence science and embedding-space alignment engineering — both of which have active prior and contemporary work. The honest way to assess uniqueness is to map that landscape precisely:

Work	What it showed	Relation to this project
Kornblith et al. 2019 (CKA)	Matched layers of separately trained vision networks are similar under CKA	Experiment A2 replicates the method on an LLM pair (GPT-2 vs Pythia) with a random-weights control
Lenc & Vedaldi 2015; Bansal, Nakkiran & Barak 2021 (model stitching)	A single linear layer connects independently trained networks with small penalty	Experiment A3 replicates closed-form stitching layer-by-layer, adding the sample-starvation diagnosis
Git Re-Basin (Ainsworth et al. 2022)	After permutation alignment, independent training runs lie in one loss basin	Theoretical grounding for why weights differ while function converges
Platonic Representation Hypothesis (Huh et al. 2024)	Representation geometries converge with model strength, across modalities	The framing hypothesis; this project is a small-scale independent confirmation
vec2vec (2025)	Translation between embedding spaces without any paired data, from geometry alone	Stronger unsupervised cousin of the adapter; also the source of the privacy caveat (embeddings are not encryption)
Text-to-Concept (Moayeri et al., ICML 2023)	Linear layers align diverse vision models' features to CLIP space, transferring CLIP's text abilities	Closest academic precedent for 'linear map into a vision-language space'; research setting, not deployment
Drift-Adapter (2025)	Lightweight adapters (Procrustes / low-rank affine / MLP) let a vector DB survive embedding-model upgrades with near-zero downtime	Closest engineering precedent: same math, applied to model migration within one tier rather than across device tiers
EmbeddingAdapters (open-source library)	Plug-and-play adapters mapping one text-embedding space into another; run cheap local models inside a premium space	Same economic motive (cheap encoder, premium space) for text RAG; not multimodal, not on-device
LlamaIndex linear query adapter	A linear transform on query embeddings improves RAG retrieval without re-embedding documents	Same 'transform one side only' asymmetry the project uses (images adapted, text untouched)

Table 1: Prior and contemporary work, and the exact relationship of each to this project.

2. What is unique here — and is it a breakthrough?

2.1 The sober answer

No single component is a world-first, and calling the project a breakthrough would overclaim: linear stitchability is published science, and adapter-based embedding translation is an emerging engineering

pattern (Drift-Adapter and the EmbeddingAdapters library are 2025 contemporaries). The accurate claim is different and still strong: **this project independently arrived at, implemented, and validated an instantiation of that pattern that the surveyed work has not covered** — and did so with unusually complete experimental hygiene.

2.2 The specific combination that appears to be novel

(a) Cross-tier, cross-vendor, multimodal pair. Surveyed systems translate between text-embedding models within one serving tier, or align research models to CLIP offline. This project bridges two *production vision encoders from different vendors* (Apple's MobileCLIP-S1 -> Google's SigLIP 2) across a *hardware boundary*: a consumer iPhone and a server sharing one live index. **(b) On-device deployment.** The adapter ships as a 0.79 MB fp16 Core ML package executing on the Neural Engine — the surveyed precedents stop at Python. **(c) End-task gating with full baselines.** Validation is not fit quality but text-to-image Recall against both a ceiling (native SigLIP) and a chance baseline (raw cross-space retrieval, $R@1 = 0.001$), with a pre-registered 90% gate. **(d) Linearity proven, not assumed.** A controlled higher-capacity MLP trained on the true objective improved cosine by +0.003 — evidence that the linear map is not merely adequate but *optimal*, i.e. the cross-space relationship is exactly linear and the residual is encoder capacity. None of the surveyed engineering work includes this control. **(e) Theory and application in one arc.** The same project replicates the convergence science (CKA + stitching + random control on an adversarially chosen LLM pair) that justifies the engineering — most work does one or the other.

2.3 Why independent convergence with 2025 work is a feature

That unrelated teams (vector-DB migration, RAG tooling) reached the same primitive in the same year is evidence the pattern is real and timely — 'embedding space as a contract, adapters at the edges' is becoming infrastructure. Arriving at it independently, from theory, with a deployment target none of them addressed (consumer mobile), is a legitimate contribution claim for a course project and a defensible seed for a write-up or open-source release.

3. Other practical use cases of the linear-map adapter

The primitive is general: *any* place where a cheap, local, or legacy encoder must interoperate with a canonical embedding space is this pattern. Concrete instances:

Use case	How the adapter is used	What it saves
Index migration / model upgrades	New encoder's vectors mapped into the legacy space (or queries mapped, documents untouched) while the corpus migrates lazily	Re-embedding a 10M-item corpus (~GPU-days) and index downtime; upgrade becomes a 2 MB deploy
Cheap-encoder economics for RAG	Local MiniLM-class model embeds locally; adapter lifts vectors into the premium (API) space; cloud paid only when chosen	Per-call embedding API fees and ~100-300 ms network latency per query
Edge-fleet ingestion	Cameras / IoT / worker nodes run tiny encoders; adapters normalize all streams into one canonical index	Central GPU capacity; raw-media bandwidth (3 KB vector vs MB-scale image)
Encoder A/B testing	Candidate encoder + adapter searched against the production index before committing	Building a parallel index per candidate
Federated / multi-team search	Teams with different embedding stacks map into one shared space via per-team matrices	Forcing a single org-wide encoder; N×N pairwise bridges (N adapters suffice)
Cross-model RAG cache reuse	Cached embeddings from a deprecated model stay queryable through an adapter	Cold-start recomputation of historical corpora
GeoFin-style pipelines	Lightweight parallel workers embed video frames/thumbnails cheaply; adapter aligns them to the analysis tier's space	Server GPU time on bulk media ingestion
Privacy-preserving indexing	Sensitive media embedded on-device; only adapted vectors leave	Uploading raw personal media (with the vec2vec caveat: embeddings leak information and are not encryption)

Table 2: Application catalog for cross-model linear adapters.

4. Does it save time, electricity, and improve local mobile use?

Yes on all three — with the honest note that the figures below are order-of-magnitude engineering estimates, stated with their assumptions, not measured on this project's hardware.

4.1 Compute and time

The adapter itself is computationally free at inference: a 512×768 matrix multiply is ~0.8 MFLOPs — roughly 1/5,000th of the MobileCLIP-S1 encoding that precedes it (single-digit GFLOPs) and vanishingly small next to SigLIP 2's tens of GFLOPs. The savings come from what it makes unnecessary: **(a)** avoided double indexing — every image encoded once instead of twice; **(b)** avoided re-encoding on migration — for a 10-million-image corpus at ~50 ms/image server-side, re-embedding costs ~140 GPU-hours; the adapter replaces it with a fit that takes *seconds* on 3,000 pairs; **(c)** avoided round-trips — on-device indexing removes the upload-encode-return loop (hundreds of ms per item, unbounded when offline).

4.2 Electricity and energy

Three mechanisms, in decreasing size: **(1) Small encoder instead of large.** MobileCLIP-S1 on the Neural Engine draws on the order of tens of millijoules per image; a server ViT-class encode on a

datacenter GPU, amortized with utilization and cooling overhead (PUE), sits around one to a few joules — one to two orders of magnitude more. The adapter lets the cheap encode *count*, so the expensive one never runs for indexing. **(2) Data movement avoided.** Wireless transmission costs real energy (order of joules per MB over cellular); shipping a 3 KB vector instead of a 2-4 MB photo cuts per-item radio energy by ~99.9%, on both the phone and the network path. **(3) One index instead of two.** Vector search energy scales with corpus count; a single shared HNSW graph halves the standing memory and search footprint versus duplicated indexes. For a personal-scale corpus the absolute savings are watt-hours; the point is architectural — the same pattern applied to fleet-scale ingestion (Section 3) scales those percentages onto real bills.

4.3 Improved local model use on mobile

This is the deepest effect: the adapter changes what a small on-device model *is*. Without it, a phone-side encoder is a silo — its vectors are useful only against other vectors it produced. With a 0.79 MB matrix, the small model becomes a first-class citizen of the large model's ecosystem: photos indexed on the beach are searchable by the server's best text encoder the moment they sync; the phone can search server content and vice versa; and the measured cost of smallness is explicit (88% of ceiling at R@1, 95-98% at R@5/@10) with a known escape hatch (server-side re-rank of top-10 restores the ceiling). It also future-proofs the device tier: swapping the phone encoder for next year's better small model requires refitting one matrix in seconds — no corpus re-index, no app architecture change. In short, adapters convert 'local model as a compromise' into 'local model as a peer with a published exchange rate.'

4.4 Limits of the savings claim

The adapter does not reduce the cost of *queries* that were already server-side text encodes; it does not compress the index (vectors are 768-d either way); its quality depends on the pair-fitting distribution (COCO-like photos here — domain shift to, e.g., documents would need refitting); and it inherits vec2vec's security lesson: adapted embeddings still carry recoverable information and must be protected like content, not like hashes.

Sources

Kornblith et al., ICML 2019 · Bansal, Nakkiran & Barak, NeurIPS 2021 · Ainsworth et al. 2022 (Git Re-Basin) · Huh et al., ICML 2024 (PRH) · Jha et al. 2025 (vec2vec, arXiv:2505.12540) · Moayeri et al., ICML 2023 (Text-to-Concept) · Drift-Adapter, arXiv:2509.23471 · EmbeddingAdapters (github.com/PotentiallyARobot/EmbeddingAdapters) · LlamaIndex linear query adapter (llamaindex.ai blog). Energy figures are engineering estimates from public per-inference and wireless-transmission energy literature; treat as order-of-magnitude.